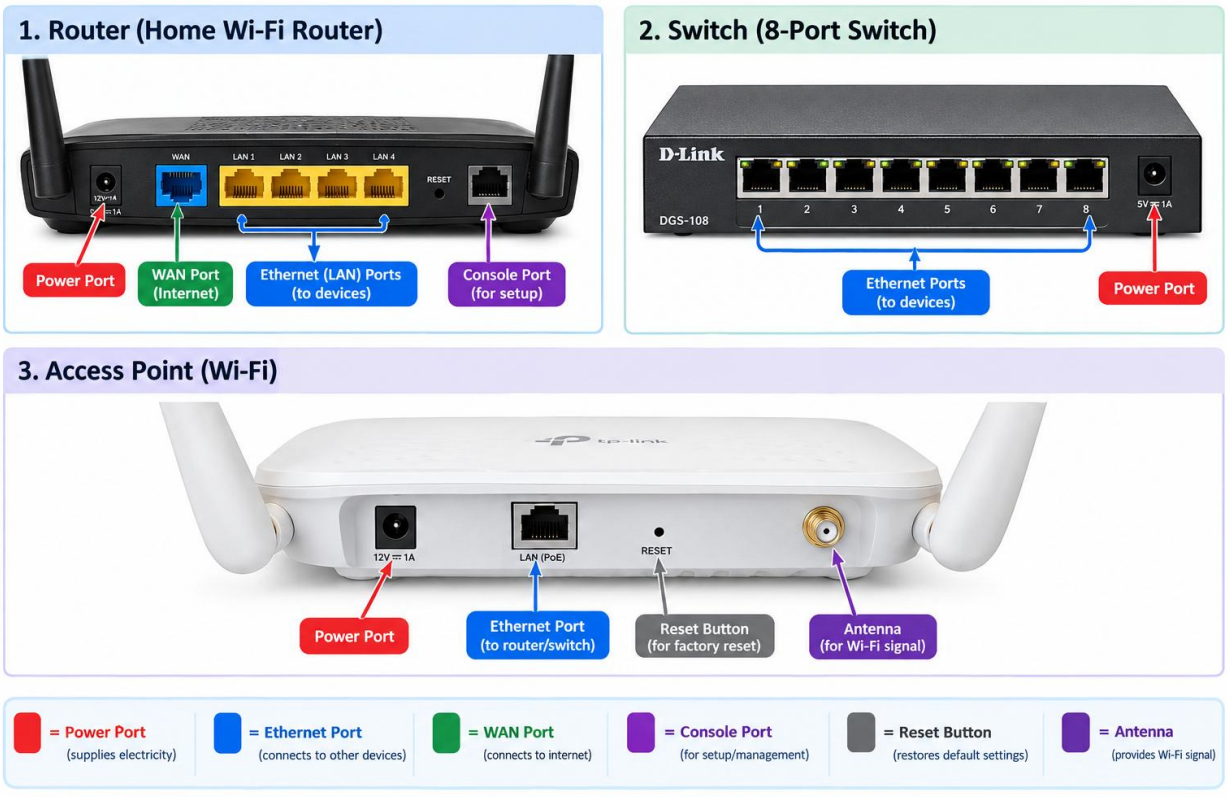
1.Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

1. **Switch: Connects computers, printers, and other wired devices.**
2. **Router: Shares the internet with devices at home.**
3. **Hub: Sends received data to every connected device.**
4. **Firewall: Helps protect devices from unwanted network access.**
5. **Access Point: Provides Wi-Fi connectivity to phones and laptops.**

2.Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



3.Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

1. **App: Instagram**
2. **When I use Instagram, my Wi-Fi router connects my phone to the internet.**
3. **The access point provides Wi-Fi so my phone can connect wirelessly.**
4. **Instagram's servers use switches to help transfer data between connected devices in their network.**
5. **A firewall helps protect Instagram's servers by blocking unwanted network traffic.**

4.Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Think about device capacity, coverage, and traffic handling.

Answer:

One possible reason is that the college Wi-Fi access point cannot handle too many students connected at the same time. This can make the internet slow and cause disconnections.

Solution: The college can upgrade to a better access point that supports more devices and provides wider Wi-Fi coverage. This can help students get a faster and more stable internet connection.